

Mariano Garcia-Melo

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EDUCATION

Loyola University Chicago

Chicago, IL

Bachelor of Science in Computer Science

Aug. 2022 – May 2026

- **Honors:** College of Arts & Sciences Dean's List (May 2024, December 2024, May 2025, December 2025, May 2026); Department of Computer Science Bonafide Computer Programmer Award (April 2023)
- **Relevant Coursework:** Data Structures I & II, Design/Analysis of Algorithms, Machine Learning, Database Programming, Operating Systems, Programming Languages, Human-Computer Interaction

EXPERIENCE

Undergraduate Research Assistant - Data Engineering

Jan. 2026 – May 2026

Loyola University Chicago

Chicago, IL

- Built ETL pipeline loading 122M+ USPTO citations across 8.6M patents into 22-table PostgreSQL database
- Wrote PyArrow streaming converter transforming 14 TSV files (largest 150M rows) to Parquet with dictionary encoding, achieving 10x compression and enabling memory-safe batch processing within an 8GB RAM ceiling
- Analyzed citation network across 638 CPC subclasses, identifying power-law behavior in 9M+ unique cited patents

Analyst

Aug. 2025 – Jan. 2026

Rambler Investment Fund

Chicago, IL

- Conducted equity research on cybersecurity SaaS company for \$1.6M student-run fund, building DCF model with sensitivity analysis to determine intrinsic value and support investment recommendation for portfolio inclusion
- Analyzed ARR, NRR, and revenue CAGR versus public cybersecurity peers using Refinitiv to assess relative valuation and support investment recommendation with comparable company analysis across key valuation metrics
- Authored "Digital Trust & Cybersecurity" section of fund's biannual report, synthesizing industry trends, competitive landscape, and regulatory developments into an investment thesis with actionable market insights

Business Systems Intern

May 2024 – Aug. 2024

IMM – Latino Factor

Arlington, TX

- Rebuilt and launched company website, driving 2,100+ unique visitors and doubling sales rep applications
- Built weighted scoring model in Excel ranking 5,000+ client accounts by priority via SAP S/4HANA data, providing consolidated prioritization sheet to logistics team for route planning
- Cleaned 50+ SAP store records, resolving missing fields and duplicates for downstream reporting accuracy

PROJECTS

OrderBook | C++, CMAKE, AWS EC2, CI/CD

Aug. 2026

- Built a C++17 limit order book processing 1M+ orders with 303ns mean / 958ns p99 latency (~3.3M ops/sec), using integer-tick pricing, price-time priority matching, and O(1) order cancellation via direct node-pointer lookups
- Replaced `std::list`'s heap allocation with a hand-built doubly-linked list and custom memory pool allocator, using pre-allocated fixed-capacity storage and free-list recycling for O(1) allocate/free, cutting a 4,000x latency outlier.
- Validated performance and correctness across AWS EC2 (shared tenancy) and dedicated hardware, with continuous integration on GitHub Actions (GCC/Linux) catching compiler-specific bugs invisible on a single dev machine.

March Madness Prediction | Python, Pandas, Scikit-learn, Parquet, CI/CD

Nov. 2025 – Jun. 2026

- Curated 2026 test dataset of 144 data points across 12 contenders, sourcing features from 3 CBB rating systems
- Built MLP neural network on 1,147 historical samples with expanding-window time-series cross-validation and GridSearchCV tuning, addressing 66:1 class imbalance via threshold tuning and preventing temporal data leakage
- Containerized pipeline with Docker and automated CI via GitHub Actions, running pytest on every push

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, HTML, CSS, Swift, JavaScript, Scala

Frameworks/Libraries: Pandas, PyArrow, NumPy, PyTest, SQLAlchemy, Scikit-learn, Matplotlib, PostgreSQL, PyTorch, MongoDB, Claude API, FastAPI, Rest APIs, MapKit, LangGraph

Developer Tools: Git, Jupyter Notebook, Firebase, VS Code, Xcode, Claude Code, Codex, Linux/Unix, GitHub Actions, Docker, Excel, AWS, Amazon Bedrock, SAP S/4HANA, Refinitiv, Tableau, Power BI

Certifications: FINRA Securities Industry Essentials (SIE) Exam (Expected Nov. 2026), Anthropic Building with the Claude API (In Progress)